

Data Compression

Unit 5

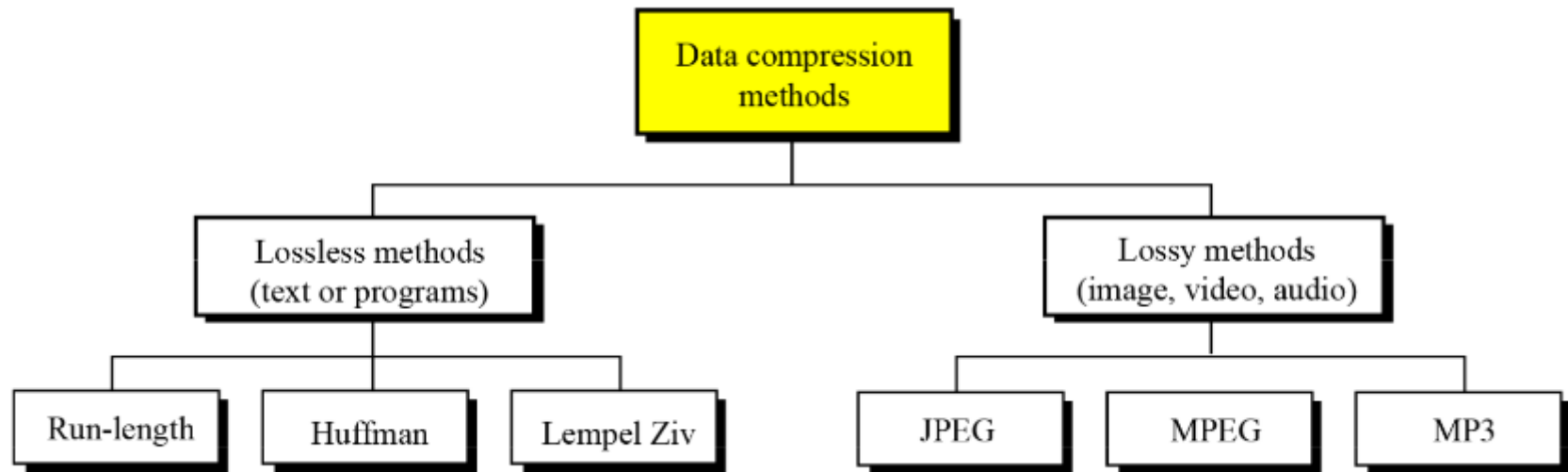
Why compress?

- To reduce the volume of data to be transmitted
- To reduce the bandwidth required for transmission and to reduce storage requirement

Data Compression

- Data Compression is the process of reducing the size of digital data to **optimize storage, transmission or processing efficiency**.
- It works by eliminating redundancy, removing irrelevant information or encoding data in most compact form while preserving its usability.

Data Compression



Storage Space

- Uncompressed graphics, audio and video data require considerable storage capacity.
- For e.g. an image represented in 640 by 480 resolutions with 8 bits per pixel require 300 kilobytes of storage space.
- So, in order to reduce the storage requirement of the multimedia data it should be compressed.
- The compression should be such that there should not be a loss of quality of the data and hence compression should not reduce the information content of the data.
- For e.g. 90% of the raw audio can be deleted without affecting the quality of the audio.

Coding Requirement

- Storage- Multimedia Elements require much more storage space than simple text.
- Based on a small window of 640 x 480 pixels on a display.
 - For the representation of the text medium, two bytes are used for every 8×8 pixel character.

$$\text{Character per screen page} = \frac{640 \times 480}{8 \times 8} = 4,800$$

$$\text{Storage required per screen page} = 4,800 \times 2 \text{ byte} = 9,600 \text{ byte} = 9.4\text{Kbyte}$$

Coding Requirement

- Individual pixels of a bitmap can be coded using 256 different colors, requiring a single byte per pixel.

Storage required per screen page = $640 \times 480 \times 1 \text{ byte} = 307,200 \text{ byte} = 300 \text{ Kbyte}$

- An uncompressed stereo audio signal of CD quality is sampled at 44.1kHz and quantized using 16 bits.

$$\text{Data rate} = 2 \times \frac{44,100}{\text{s}} \times \frac{16 \text{ bit}}{8 \text{ bit/byte}} = 176,400 \text{ byte/s}$$

$$\text{Required storage space/s} = 176,400 \text{ byte/s} \times \frac{1 \text{ s}}{1,024 \text{ byte/Kbyte}} = 172 \text{ Kbyte}$$

Coding Requirement

- These examples briefly show the increased demands on a computer system in terms of required storage space and data throughput if still images and in particular continuous media are to be processed
- Multimedia systems is subject to certain constraints.
 - The quality of the compressed data should be as good as possible.
 - To make cost-effective implementation possible, the complexity of the technique used should be minimal.
 - The processing of the algorithm must not exceed certain time span.

Entropy Coding

- Entropy coding can be used for different media regardless of the medium's specific characteristics.
- The data to be compressed are viewed as a sequence of digital data values, and their semantics are ignored.
- It is lossless because the data prior to encoding is identical to the data after decoding; no information is lost.

Huffman Encoding

- David Huffman in 1951
- Encoding follows the prefix rule
- Most generated char will get small code and least generated char will get larger code.
- Time complexity: $\theta(n \log n)$

Prefix Rule: No code is prefix of another

Huffman Encoding

We have solved the examples in note copy

E.g let the word to be encoded be AABABBEFFFFEFC

Run Length Encoding

- Run Length coding is very simple form of data compression in which runs of data (that is sequences in which the same data value occurs in many consecutive elements) are stored as single data value and count rather than as the original run.
- **Run Length Encoding Format:** Value, Count

Run Length Encoding

Example

1. Given string: aaaabbbccccccddd

RLE format: a4b2c5d3

1. Given string: abcd

RLE format: a1b1c1d1

Run Length encoding

Example

1. Given string: 0001111100011

RLE format: (0,3) (1,5) (0,3) (1,2)

1. Given string: 010101

RLE format: (0,1) (1,1) (0,1) (1,1) (0,1) (1,1)

Run Length encoding

- Drawback

If there are no sequences in the data RLE doubles the amount of number compared with original data.

- Advantages

- RLE is reversible / lossless compression
- RLE is particularly well suited to bitmap images

Arithmetic encoding

- Unlike Huffman coding which used a separate codeword for each character, arithmetic coding yields a single codeword for each encoded string of characters.
- The first step is to divide the numeric range from 0 to 1 into a number of different characters present in the message to be sent including the termination character – and the size of each segment by the probability of the related character

Source Encoding

- Source coding takes into account the semantics of the information to be encoded
- In the case of lossy compression, a relation exists between the uncoded data and the decoded data; the data streams are similar but not identical.
- For example: In audio, the melody, rhythm, and overall sound quality are preserved but High-frequency sounds in audio might be removed or reduced in quality.

Source Encoding

- In the case of speech, a considerable reduction of the amount of data can be achieved by transforming the time-dependent signal into the frequency domain, followed by an encoding of the formants

Hybrid Encoding

- This type of coding mechanism involve the combine use of both the source coding and the entropy coding for enhancing the compression ratio still preserving the quality of information content.
- The example of Hybrid Coding includes MPEG, JPEG, H.261, DVI techniques.

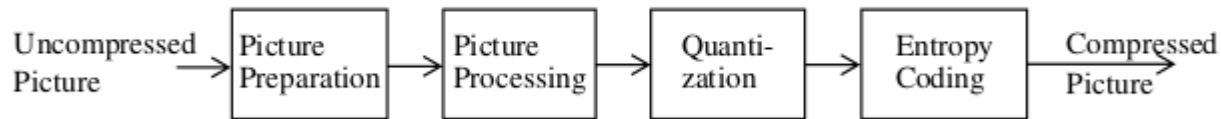


Figure: Image Compression with hybrid encoding

Hybrid Encoding

Step 1:

Goal - Remove perceptually insignificant data.

Methods

- **Images:** Discrete Cosine Transform (DCT) in JPEG, quantization.
- **Audio:** Psychoacoustic masking, formant/pitch modeling .
- **Video:** Motion compensation, spatial/temporal redundancy removal.

Hybrid Encoding

Step 2:

Goal - Further compress the residual data without quality loss.

Methods

- Entropy coding (Huffman, Arithmetic coding).
- Run-Length Encoding (RLE) or Lempel-Ziv algorithms (LZ77, LZ78).

Lossy Compression Methods

Our eyes and ears cannot distinguish subtle changes. In such cases, we can use a lossy data compression method. These methods are cheaper—they take less time and space when it comes to sending millions of bits per second for images and video. Several methods have been developed using lossy compression techniques. **JPEG (Joint Photographic Experts Group)** encoding is used to compress pictures and graphics, **MPEG (Moving Picture Experts Group)** encoding is used to compress video, and **MP3 (MPEG audio layer 3)** for audio compression.

Image compression – JPEG encoding

an image can be represented by a two-dimensional array (table) of picture elements (pixels).

A grayscale picture of 307,200 pixels is represented by 2,457,600 bits, and a color picture is represented by 7,372,800 bits.

In JPEG, a grayscale picture is divided into blocks of 8×8 pixel blocks to decrease the number of calculations because, as we will see shortly, the number of mathematical operations for each picture is the square of the number of units.

Image compression – JPEG encoding

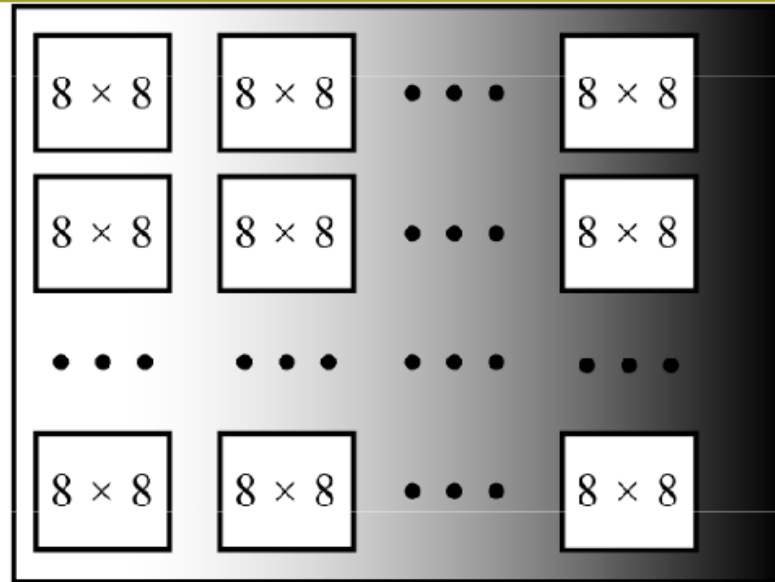


Figure 2 JPEG grayscale example, 640 × 480 pixels

Image Compression – JPEG encoding

The whole idea of JPEG is to change the picture into a linear (vector) set of numbers that reveals the redundancies. The redundancies (lack of changes) can then be removed using one of the lossless compression methods we studied previously. A simplified version of the process is shown in Figure 3.

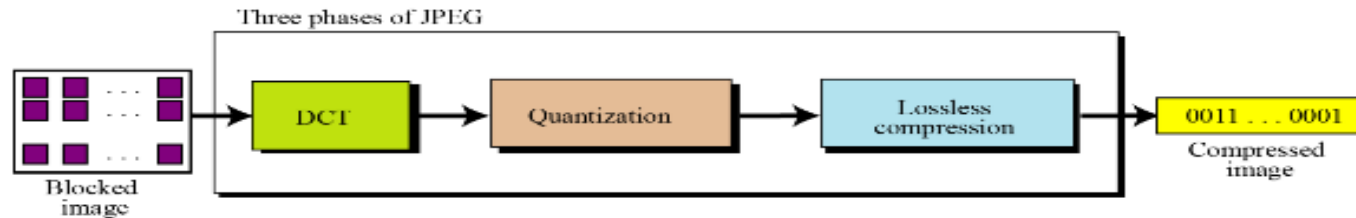


Figure 3 The JPEG compression process

Discrete Cosine Transform(DCT)

Why Discrete Cosine Transform is used in image compression?

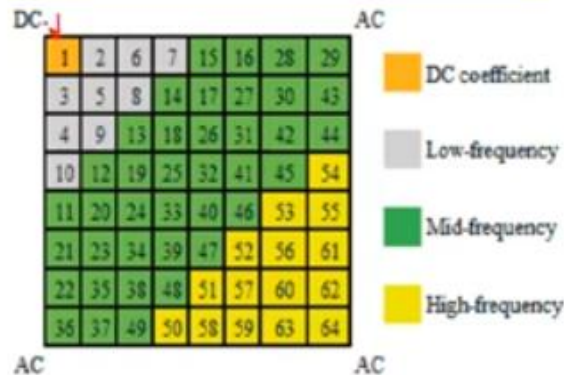
- Discrete Cosine Transform is used in lossy image compression
- The DCT works by separating images into parts of different frequencies
- The image is broken into 8 x 8 blocks of pixels
- During quantization, the less important frequencies are discarded hence the use of term lossy
- i.e its large amount of information is stored in very low frequency component of a signal and rest other frequency having very small data which can be stored by using very less number of bits.

DCT

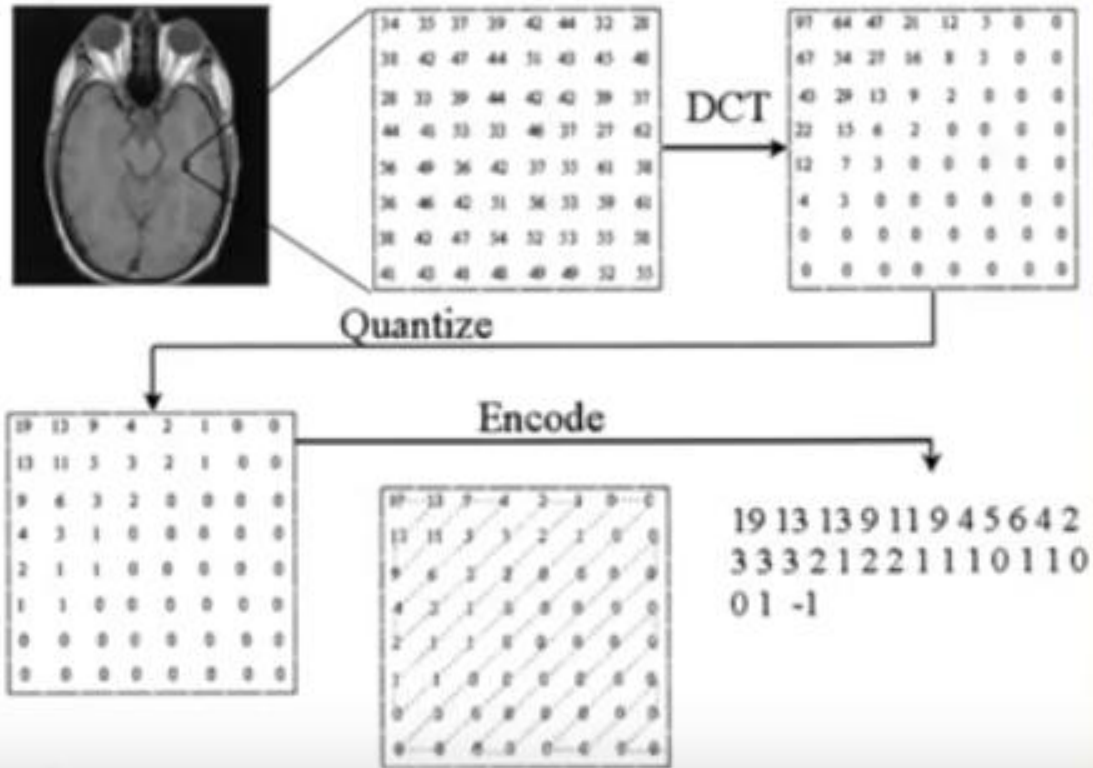
The general equation for a 2D (N by M image) DCT is defined by the following equation:

$$F(u, v) = \left(\frac{2}{N}\right)^{\frac{1}{2}} \left(\frac{2}{M}\right)^{\frac{1}{2}} \sum_{i=0}^{N-1} \sum_{j=0}^{M-1} \Lambda(i) \cdot \Lambda(j) \cdot \cos \left[\frac{\pi \cdot u}{2 \cdot N} (2i + 1) \right] \cos \left[\frac{\pi \cdot v}{2 \cdot M} (2j + 1) \right] \cdot f(i, j)$$

Distribution of coefficients in DCT



DCT



Lossy Sequential DCT based Mode

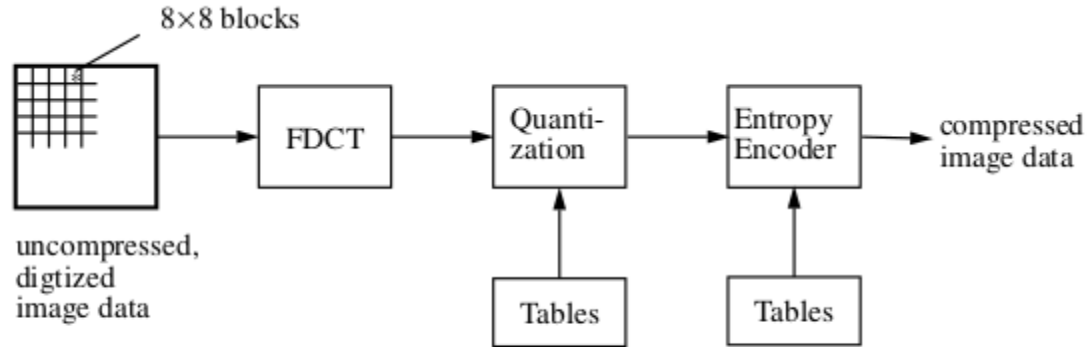


Figure 7-9 Steps of the lossy sequential DCT-based coding mode, starting with an uncompressed image after image preparation.

Lossy Sequential DCT based Mode

Image Processing

- It basically involves the block preparation where the image samples are grouped into 8×8 pixels and passed to the encoder.
- Then Discrete Cosine Transformation is applied to the blocks
- Each of these values is then transformed using Forward DCT (FDCT).

Lossy Sequential DCT based Mode

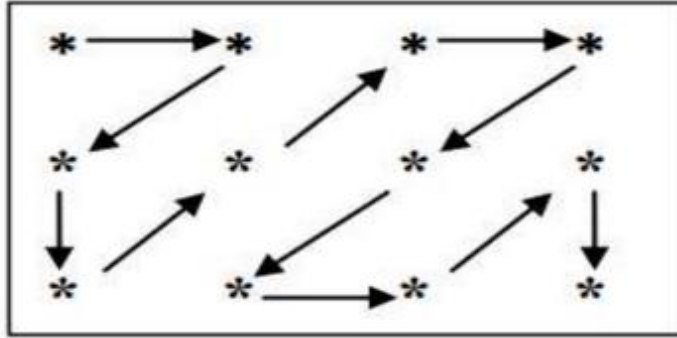
Quantization

- The JPEG application provides a table of 64 entries. Each entry will be used for the quantization of one of the 64 DCT-coefficients.

Entropy Encoding

- During the initial step of entropy encoding, the quantized DC-coefficients are treated separately from the quantized AC-coefficients.
- The DCT processing order of the AC coefficients involves the zigzag sequence to concentrate the number of zeros

Lossy Sequential DCT based Mode



In this lossy sequential DCT-based mode, only Huffman encoding is allowed. In lossy sequential mode the framework of the whole picture is not formed but parts of it are drawn i.e. sequentially done.

Expanded Lossy DCT based Mode

- It differs from the sequential mode in terms of number of bits per sample. Here 12 bits along with 8 bits per sample can be used.
- It specifies Huffman and arithmetic encoding as entropy encoding methods.
- For the expanded lossy DCT-based mode, JPEG specifies progressive encoding in addition to sequential encoding.
- At first, a very rough representation of the image appears which is progressively refined until the whole image is formed. This progressive coding is achieved by layered coding.

Expanded Lossy DCT based Mode

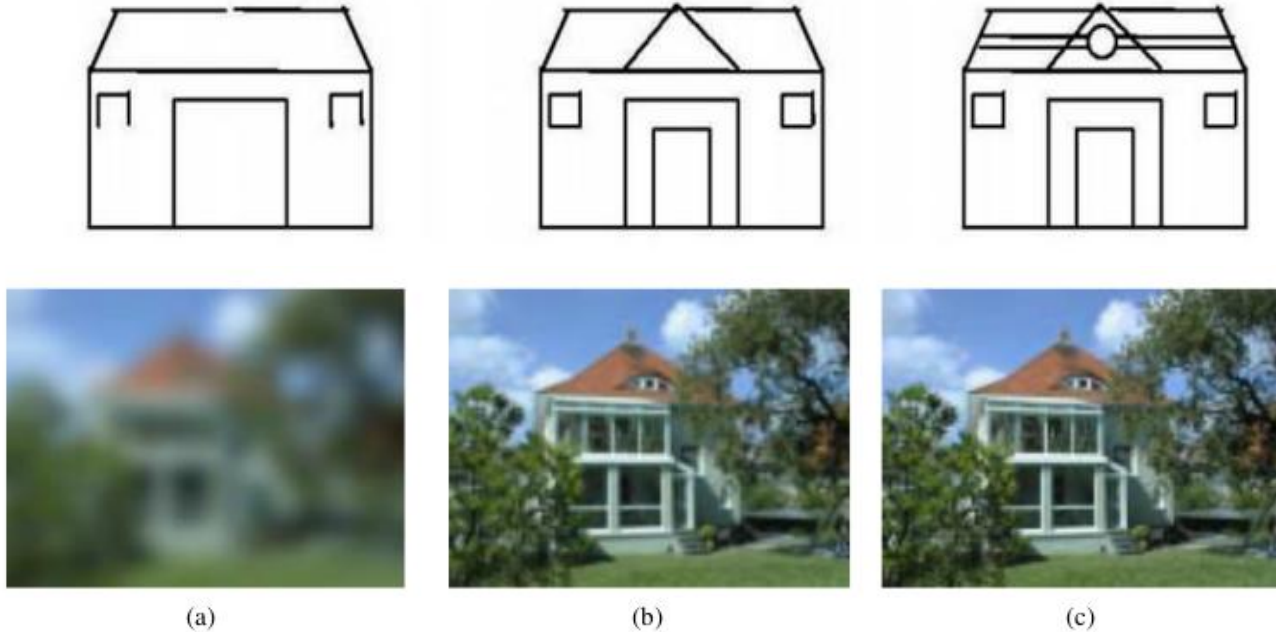


Figure 7-16 Progressive picture presentation: (a) first phase, very unsharp image; (b) second phase, unsharp image; (c) third phase, sharp image.

JPEG Compression

- Youtube Video to Watch
- https://www.youtube.com/watch?v=Kv1Hiv3ox8I&ab_channel=BranchEducation

MPEG Compression

- The Motion Pictures Expert Group was formed by the ISO to formulate a set of standards relating to a range of multimedia applications that involve the use of video with sound.
- MPEG can deliver a data rate of at most 1856000 bits/second, which should not be exceeded.
- Data rates for audio are between 32 and 448 Kbits/second; this data rate enables video and audio compression of acceptable quality.

MPEG Compression

The MPEG standard exploits the other standards and they are

- **JPEG**

JPEG is used as the motion picture is a continuous sequence of still image.

- **H.261**

Since the H.261 standard already existed during the MPEG work, the MPEG group strived to achieve at least a certain compatibility between the two standards in some areas.

The structured working of MPEG Video compression

Spatial Redundancy

Within a single frame, adjacent pixels often correlate. MPEG uses **intra-frame compression** (similar to JPEG)

- **Macroblocks:** Frames are divided into blocks (e.g., 16x16 pixels).
- **Discrete Cosine Transform (DCT):** Converts spatial data into frequency coefficients, concentrating energy into fewer components.
- **Quantization:** Coefficients are approximated (lossy step), prioritizing perceptually significant data.
- **Entropy Encoding:** Huffman or arithmetic coding compresses quantized data.

Temporal Redundancy

Extreme case: stationary video



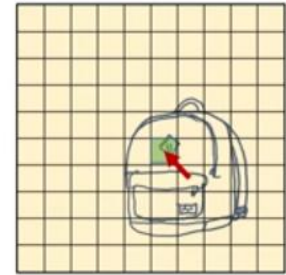
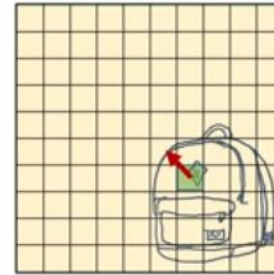
repeat(



, 4)

Temporal Redundancy

Block motion estimation



A little more realistic case: some parts change



Temporal Redundancy

Between consecutive frames, much content remains static. MPEG uses inter-frame compression:

- **Motion Estimation:** Tracks movement of macroblocks across frames, generating motion vectors.
- **Motion Compensation:** Predicts frames using reference frames (I/P-frames) and motion vectors, encoding only residuals (differences).

Type of Image frames in MPEG

- There are two basic types of compressed frame: those that are encoded independently and those that are predicted.
- The first are as intra coded frames or I-frames.
- In practice, there are two types of predicted frames: predictive or P-frames and bidirectional or B-frames

I-frames

- I-frames are encoded without reference to any other frames. Each frame is treated as a separate (digitized) picture and the Y, Cb, and Cr matrices are encoded independently using the JPEG algorithm.
- The level of compression obtained with I-frames is relatively small.
- I-frames must be present in the output stream at regular intervals in order to prevent for the possibility of the contents of an encoded I-frame being corrupted during transmission.
- The number of frames/pictures between successive I-frames is known as a group of pictures or GOP.

P-frames

- The encoding of a P-frame is relative to the contents of either a preceding I-frame or a preceding P-frame.
- As indicated, P-frames are encoded using a combination of motion estimation and motion compensation and hence significantly higher levels of compression can be obtained.
- The number of frames between a P-frame and the immediately preceding I-frame is called the prediction span.

B-Frames

- A Bidirectional Predictive Frame that uses both past and future reference frames (I or P-frames) for prediction.
- Achieves higher compression by leveraging motion in both directions.
- Requires reordering during encoding/decoding (future frames must be processed first).

Type of Image frames in MPEG

